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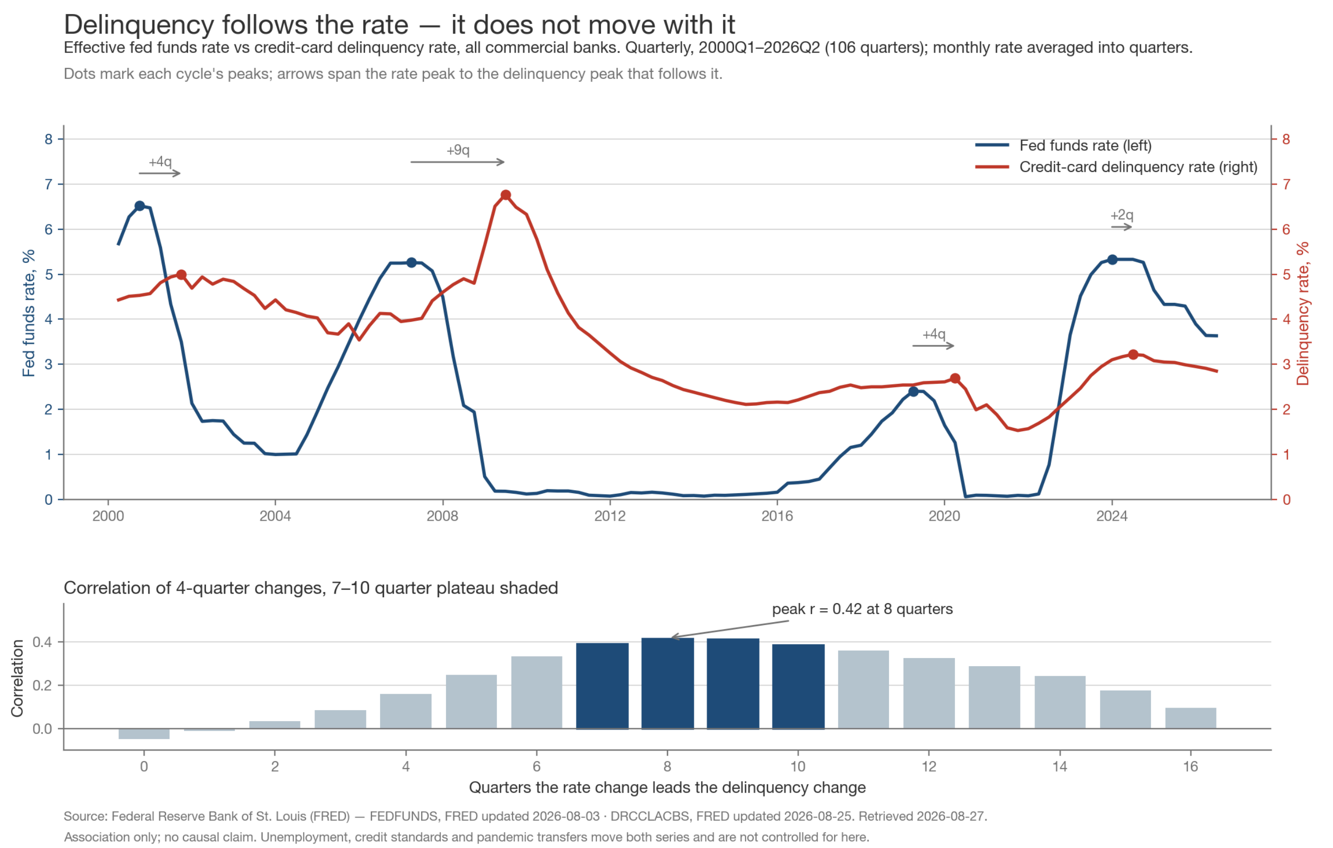
Rate changes reach the card book about two years later



Fed funds rate and credit-card delinquency, 2000Q1–2026Q2, with the cross-correlation of four-quarter changes peaking at an eight-quarter lag

The finding

Credit-card delinquency does not move with the policy rate. It moves **after** it, by roughly two years.

Correlating four-quarter changes in the effective fed funds rate against four-quarter changes in the credit-card delinquency rate, the correlation is **0.42 at an eight-quarter lag** and **–0.05 at zero lag** — the two series are essentially unrelated in the same quarter and most related two years apart. The peak is a plateau, not a point: correlation stays above 0.37 across a **7-to-10 quarter** window, so the honest statement of the lag is a range, not a date.

The relationship is stronger once the pandemic is set aside, when fiscal transfers pushed delinquency to a sample low of 1.53% (2021Q3) while the rate sat at zero. Excluding 2020–21 the peak correlation is **0.56 at nine quarters**; on pre-2020 data only it is **0.71 at nine quarters**.

It is a timing signal, not a forecast. At the eight-quarter lag a one-point four-quarter move in the rate maps to **0.18 points** of delinquency change, and the fit explains **17% of the variance** over the full sample (31% excluding 2020–21). Rate direction tells a risk team roughly *when* to expect pressure — not how much.

Results

Data. Two FRED series since 2000-01-01: `FEDFUNDS` (Federal Funds Effective Rate, monthly) and `DRCCLACBS` (Delinquency Rate on Credit Card Loans, All Commercial Banks, quarterly). Retrieved 2026-08-27; FRED last updated `FEDFUNDS` 2026-08-03 and `DRCCLACBS` 2026-08-25.

Cleaning and alignment. The monthly rate was averaged into quarters — a quarter's rate is the mean of its three months, which is what a rate comparison means; sampling the last month instead would discard two thirds of the observations. The join is inner, so every row is complete: **106 quarters, 2000Q1–2026Q2, no gaps and no missing values**. 2026Q3 is dropped because it is partial — `FEDFUNDS` reports through July 2026 and `DRCCLACBS` through 2026Q2, and the delinquency series runs roughly a quarter behind by design.

Lag estimate. Both series are strongly trending and highly autocorrelated, so a correlation in levels would mostly measure the shared business cycle. Differencing to change removes that. Two differencing choices agree: quarter-over-quarter change peaks at nine quarters (r = 0.29), four-quarter change at eight (r = 0.42). The four-quarter version is the one charted — same question, less reporting noise.

CHECK	PEAK LAG	R
4-quarter change, full sample (n = 94)	8 quarters	0.42
4-quarter change, excluding 2020–21	9 quarters	0.56
4-quarter change, pre-2020 only	9 quarters	0.71
Quarter-over-quarter change, full sample	9 quarters	0.29

Cycle-by-cycle cross-check. Measuring peak to peak instead — the quarters between each cycle's rate top and the delinquency top that follows it — gives **+4, +9, +4 and +2 quarters** for the 2000, 2007, 2019 and 2023 cycles. That is noisier than the correlation estimate and rests on four observations, and it picks a local top rather than the whole change path. The two methods agree on the ordering — delinquency follows — and disagree on the magnitude. Treat the lag as **one to two and a half years**, centred near two.

Where the current cycle sits. The rate peaked at **5.33% in 2023Q4**; delinquency peaked at **3.22% in 2024Q2** and has fallen to **2.85%** as of 2026Q2, against a 2000–2019 average of 3.72%. The rate is now 3.63%, down 1.70 points from its peak. Eight quarters of rate cuts (four-quarter changes of –0.68 to –1.00 points, 2024Q4 through 2026Q2) have not yet landed; on the fitted slope they imply a further **0.12 to 0.18 points** of four-quarter delinquency decline arriving between 2026Q3 and 2028Q2.

What this is not. Association, not causation, and univariate. Unemployment, credit standards, card originations and the 2020–21 transfers all move both series and none are controlled for here. The overlapping four-quarter windows also mean the effective sample is well below 94, so no p-value is quoted; the robustness columns above are the evidence, and the breadth of the plateau is the uncertainty.

What follows for a risk team

1. **Read rate direction as a two-year early-warning input, not a within-year one.** Nothing in this data supports expecting the card book to respond to a rate move inside four quarters — at lags of zero to three quarters the correlation is indistinguishable from zero.
2. **Plan the stress window as a window.** The 7-to-10 quarter plateau means a stress scenario dated to a specific quarter is over-precise. Provision and staffing plans should span the range.
3. **Do not use this alone.** Seventeen percent of variance explained is a timing prior, not a model. Pair rate direction with unemployment and with origination-vintage quality, which is where the size of the loss actually comes from.
4. **The current read is benign, and conditional.** Delinquency has been falling since 2024Q2 and the easing already in the pipe points the same way through
5. That inference holds only while employment does. The one stretch in this sample where the relationship broke — 2020–21 — broke because something outside the rate moved consumers' ability to pay, and a labour-market shock would break it the same way, in the other direction.
6. **Re-run on the quarterly release.** `DRCCLACBS` lands about a quarter in arrears; the next observation moves the last point on this chart, and the fetch, alignment and lag estimate here re-run unchanged against it.

Analysis run 2026-08-27 against FRED data retrieved the same day. Source: Federal Reserve Bank of St. Louis (FRED), series `FEDFUNDS` and `DRCCLACBS`.
